ZZ-B-586D
December 15, 1976
SUPERSEDING
(See Section 6)

FEDERAL SPECIFICATION

BAG, ICE AND HOT WATER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers ice bags and hot water-ice bags for use in medical facilities.

1.2 Classification. This specification covers bags of the following types:

Type I - Ice bag, oval.
Type II - Ice bag, throat type.
Type III - Hot water-ice bag, stopperless.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

FSC 6530

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Federal Standards:

Fed. Test Method Std. No. 601 - Rubber: Sampling and Testing.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 412 - Tension Testing of Vulcanized Rubber.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber. The rubber used shall be natural rubber, synthetic rubber or a suitable mixture of synthetic and natural rubber. The rubber compound shall be free from any ingredient that might cause irritation of the skin. The rubber material shall have the following physical properties when tested as specified herein.

TABLE I. Physical properties of rubber

Physical property	Requirement	Test procedure paragraph
Tensile strength	1500 pounds per square inch minimum --------	4.4.1
Ultimate elongation	500 percent minimum --	4.4.2
Resistance to accelerated aging		
Tensile strength	25 percent maximum decrease ----------	4.4.3
Ultimate elongation	25 percent maximum decrease ----------	4.4.3
Resistance to boiling water (for class III only)		
Tensile strength ----	50 percent maximum decrease ----------	4.4.5
Ultimate elongation --	50 percent maximum decrease ----------	4.4.5

3.1.2 Metal plate. The metal plate used in the type III bag shall be fabricated from corrosion-resisting metal, or shall be rendered corrosion resisting by suitable plating.

3.1.3 Filler cap collar. The filler cap collar shall be made of corrosion-resisting steel, shall be made corrosion-resisting by nickel plating, or shall be made of aluminum.

3.1.4 Filler cap. The filler cap shall be made of corrosion-resisting metal, polystyrene, polyethylene, or polypropylene plastic.

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3.1.5 Tying tabs. The tying tabs shall be made of double-layered gum rubber, rubberized fabric, or molded rubber as specified herein.

3.2 Design and construction. Bags shall be as specified herein. Variations in contours or thicknesses indicated in the figures, where not limited by dimensional requirements, shall be acceptable. The exterior surfaces of the body of the bag shall be smooth for the patient's comfort. Bags shall be any standard commercial color.

3.2.1 Type I bag. The type I bag shall conform to figure 1 and as specified herein. Bag shall have walls reinforced at seams and throat. Bag shall be closed with a screw cap and shall be watertight.

3.2.1.1 Filler opening. The filler opening shall be a minimum of 2 inches in diameter.

3.2.1.2 Filler cap. The cap shall be threaded to mate with the thread in the filler opening. The cap shall be furnished with a suitable sealing washer, securely held in place.

3.2.1.3 Tying tabs. Bag shall have two or four tabs for tying (see 3.1.5). If tabs are made of molded rubber, they shall have a reinforced hole. Two of the tabs shall be secured to the edge of the bag at the opposite ends of the long axis. If four tabs are provided, they shall be equidistantly spaced around the periphery.

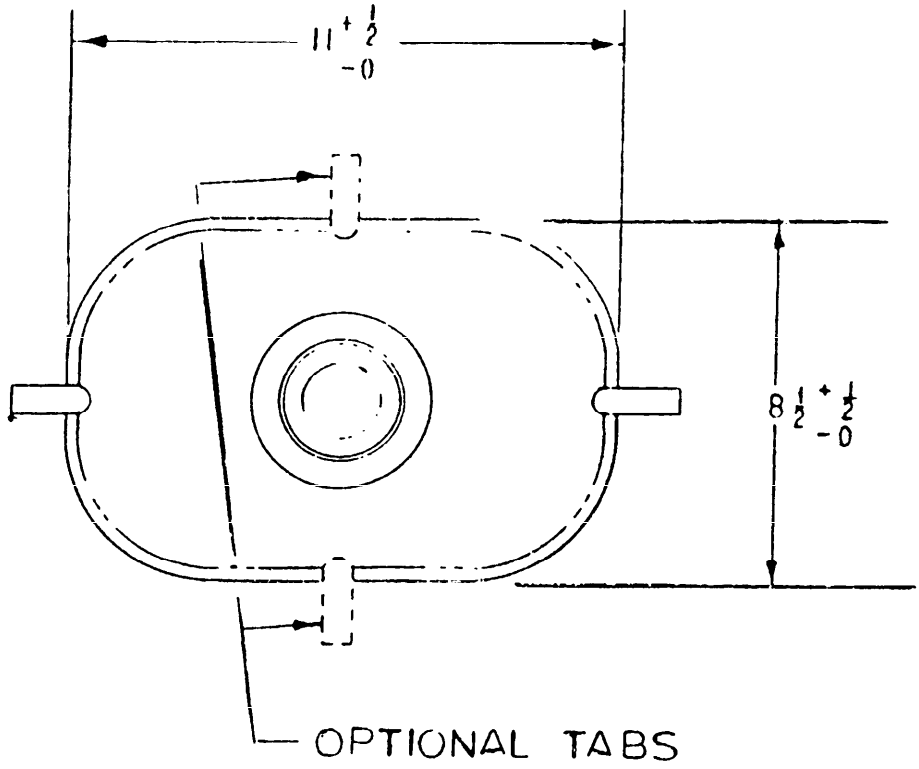
3.2.1.4 Resistance to internal pressure. The bags shall not leak or show any indication of weakness, and the weight of the bag and water shall not increase more than 5 percent, when tested as described in 4.4.6.1.

3.2.2 Type II bag. The type II bag shall conform to figure 2 and as specified herein. Bag shall be reinforced at seams and throat. Bag shall be closed with a filler cap and shall be watertight. The sides shall be concave, so that the width in the center shall be 3/4 $\pm$ 1/8 inch less than the width at the ends.

3.2.2.1 Filler opening. The filler opening shall be not less than 2 nor more than 2-1/2 inches in diameter and shall be located near one end.

3.2.2.2 Filler cap. The cap shall be threaded to mate with the filler opening. The cap shall be furnished with a suitable sealing washer, securely held in place.

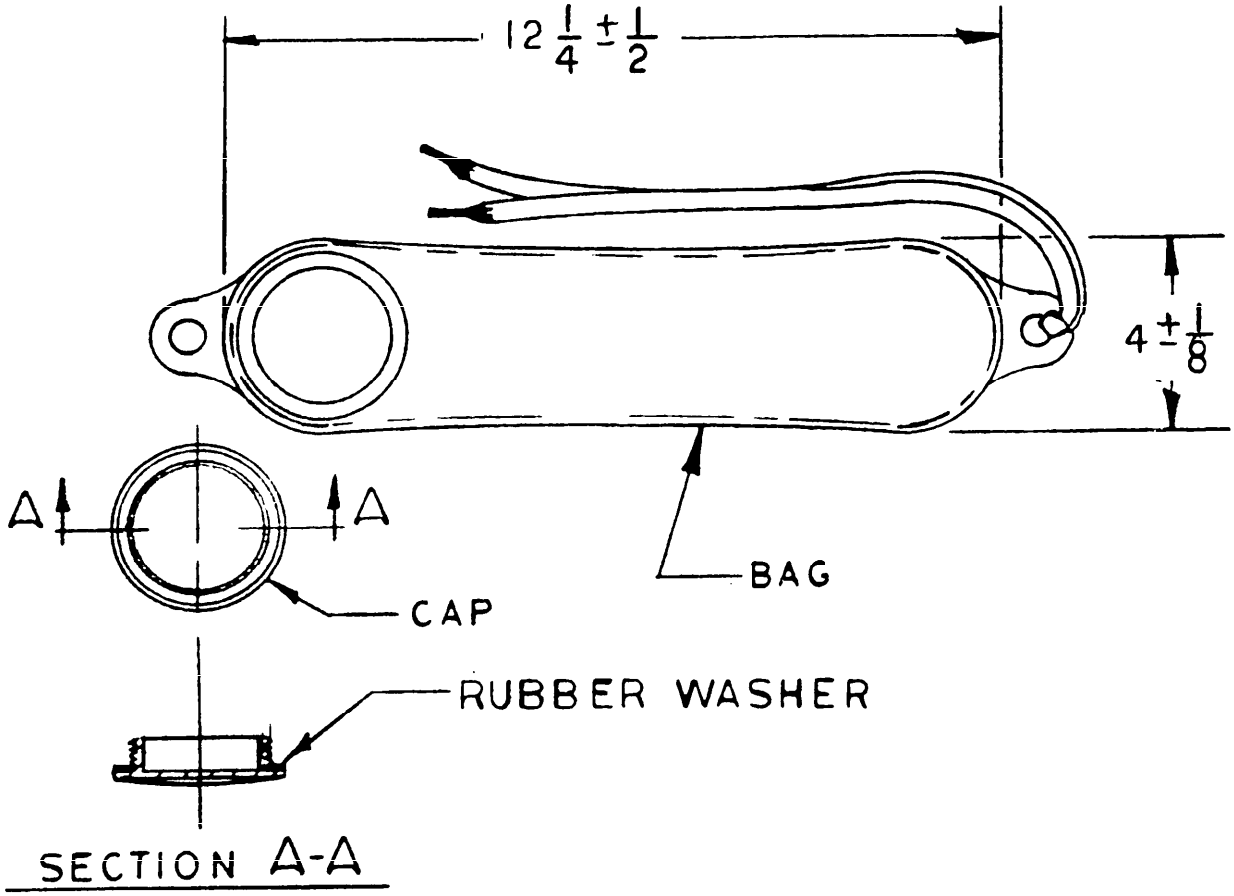
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NOTE: All Dimensions are in inches.

FIGURE 1. Ice bag, oval.

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NOTE: All Dimensions are in inches.

FIGURE 2. Ice bag, throat.

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3.2.2.3 Tying tabs. The bag shall have two tabs for tying. Tabs shall be made of rubberized fabric or molded rubber, and each tab shall have a reinforced hole. A suitable tie tape shall be provided with each bag. The tape shall be long enough to tie the bag securely, using a bow knot, around a cylinder 20 inches in circumference.

3.2.2.4 Resistance to internal pressure. The bag shall show no leakage nor any indication of weakness and the increase in weight of the bag shall not exceed 5 percent when subjected to the hydrostatic-pressure test described in 4.4.6.1.

3.2.3 Type III bag. The bag and metal plate shall be in accordance with figure 3 for style and design. The bag shall have a minimum capacity of 2 quarts of water. The bag shall be suitable for use either as a hot water bottle or as an ice bag as required. The neck of the bag shall function as a funnel in filling the bag. In filling the bag, the locking tabs shall be unbuttoned and the metal plate positioned toward the ends of the lower tab. The neck shall be manually opened and held open for filling. After filling, the neck shall be closed by pushing the metal plate back against the neck and buttoning the short tabs; and then folding the neck twice over itself and buttoning the long tabs. When tested as specified in 4.4.6.2, the bag shall show no evidence of rupture, ballooning, or leakage; and the loss of water through the closure shall not be more than 5 percent. When tested as specified in 4.4.7, the tab on bottom of the bag shall have sufficient strength and rigidity to support a filled bag without damage to the eye of the tab. In addition, the bag shall show no evidence of leakage through the closure. The bag shall show no evidence of leakage when tested as specified in 4.4.8.

3.3 Marking. Each bag shall be marked, legibly and permanently, with the manufacturer's name or registered trademark. Marking shall not interfere with the function nor detract from the appearance of the bag. In addition, each type III bag shall have permanent and legible instructions for closure on the neck.

3.4 Colorfastness. The color shall not rub off the bag when the bag is tested as specified in 4.4.4.

3.5 Instructions. Printed instructions for use shall be included with each type III bag.

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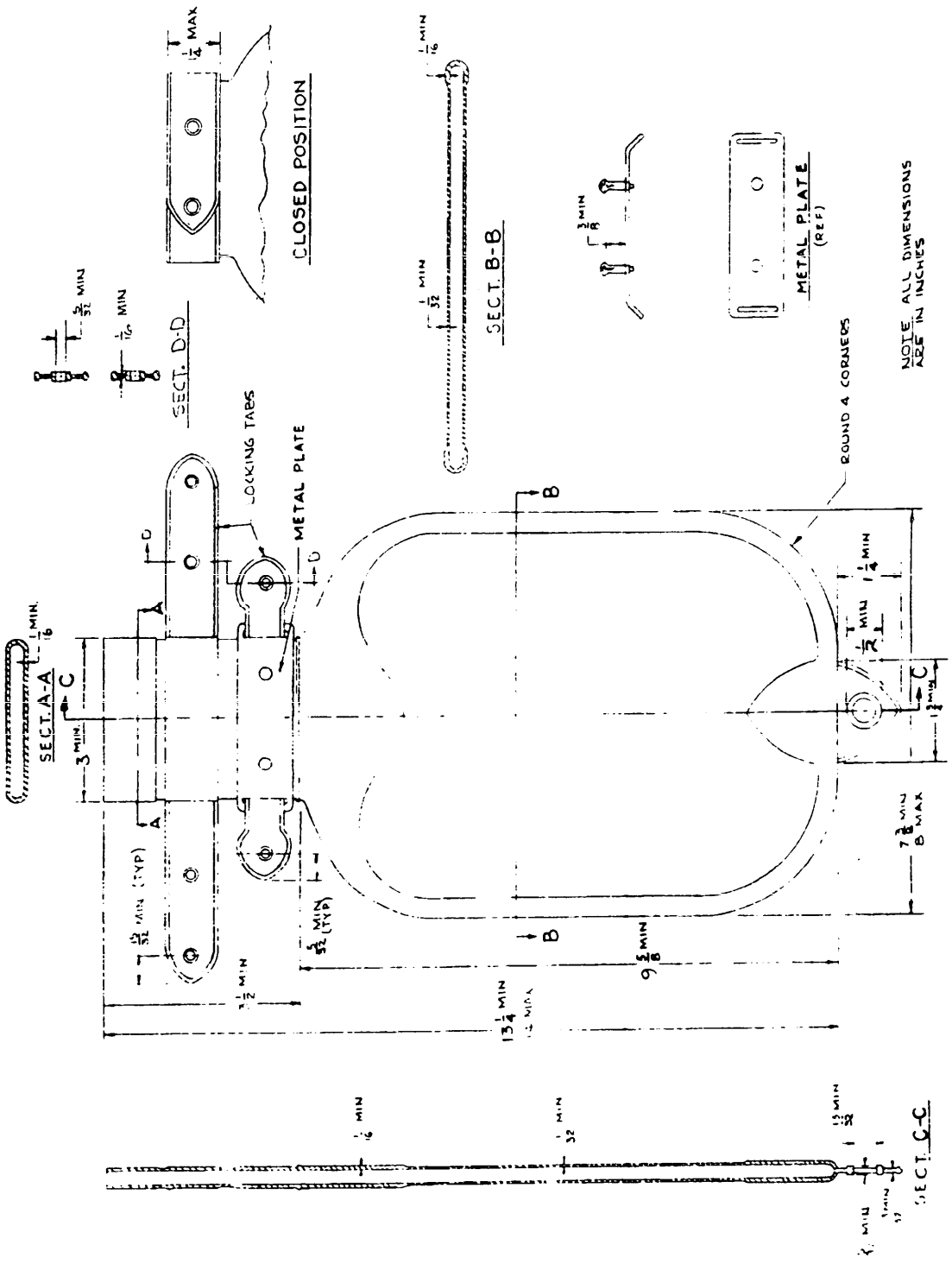


FIGURE 3. Ho[illegible]ater-ice bag.

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3.6 Workmanship. Workmanship shall be first class throughout. The bag shall be free from defects which detract from its appearance or may impair its serviceability. Bag shall be free of tackiness, and inner surfaces of bag shall not adhere to each other. The bag shall drain completely. All exterior surfaces shall be free of pits, cracks, cuts, pinholes, and other imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table II.

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TABLE II. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	I	2.5
Minor defects	I	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table III. Inspection level shall be S-1 and an acceptance level of zero shall apply for all tests.

TABLE III. Sampling criteria for tests

Component (Unit of product)	Characteristic	Requirement	Test procedure
Bag material (all models)	Tensile strength	3.1.1	4.4.1
	Ultimate elongation	3.1.1	4.4.2
	Resistance to aging	3.1.1	4.4.3
	Permanency of color	3.4	4.4.4
Type III bag material	Resistance to boiling water	3.1.1	4.4.5
Type I bag and Type II bag	Resistance to internal pressure	3.2.1.4 and 3.2.2.4	4.4.6.1
Type III Bag	Resistance to internal pressure	3.2.3	4.4.6.2
	Load strength of tab	3.2.3	4.4.7
	Leakage	3.2.3	4.4.8

4.3 Examination. The bags shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table IV. Examination shall not be limited to the classified possible defects listed below.

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TABLE IV. Classification of defects

Categories	Defects
Major:	All types
101	Bag not designed and constructed as specified.
102	Component(s) missing.
103	Surface not free of tackiness.
104	Inside surfaces stick together.
105	Surface not free of cuts, cracks, abrasions, tears, pits, and pinholes.
106	Eyelets not free of cuts, breaks, flash, and sharpness.
	Types I and II
107	Tying tab missing or not located as specified.
108	Rubber tab hole not reinforced.
109	Thread of screw cap does not mate smoothly with thread of opening on bag.
110	Tie tape not long enough. (type II.)
	Type III
111	Metal plate not free of corrosion, peeling, flaking, and chipping.
112	Metal plate not free of burrs and sharp edges.
113	Closure instructions not present, properly located, permanent and legible.
Minor:	All types
201	Surface not free of unspecified protrusions, occlusions, flash, and imbedded foreign matter.
202	Bag does not drain completely.
203	Bag not free of dirt, grease, and foreign matter.
204	Identification marking missing or not as specified.
205	Instructions for use not included with each type III bag.

4.3.2 Dimensional examination. The bag shall be examined for defects in dimensions. Any dimension not within the tolerances specified shall be classified as a defect.

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4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE V. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests.

4.4.1 Tensile strength. The tensile strength of the rubber compound shall be determined as described in method 4111 of Fed. Test Method Std. No. 601.

4.4.2 Ultimate elongation. Ultimate elongation of the rubber compound shall be determined as described in ASTM D 412.

4.4.3 Resistance to accelerated aging. The resistance of the rubber compound to accelerated aging in circulating air shall be determined as described in method 7221 or 7231 of Fed. Test Method Std. No. 601. The resistance to aging shall be measured by the change in tensile strength and ultimate elongation.

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4.4.4 Permanency of color. The surface of the body of the bag shall be rubbed briskly with a ball of soft cotton for a period of one minute. The surface of the bag and cotton ball shall be inspected for any evidence of color rubbing off the bag.

4.4.5 Resistance to boiling water. The resistance of the rubber compound of the type III bag to boiling water shall be determined as described in method 6611 of Fed. Test Method Std. No. 601.

4.4.6 Resistance to internal pressure.

4.4.6.1 Types I and II. The bag, shall be completely filled with water and stoppered while it is suspended at the collar. The weight shall then be determined. A two-way stopper, tightly fitted into the neck of the bag, shall be connected to a source of water supply and to a manometer tube. All air shall be removed from the system and water forced into the bag until it rises in the manometer tube to a point 36 inches above the horizontal surface on which the bag rests. During the next 30 minutes, water shall be added to the system as may be necessary to maintain the same manometer reading and the bag examined for leakage and any indication of weakness. The bag shall then be filled as described in the first sentence of this paragraph, weighed, and the percentage increase in weight calculated.

4.4.6.2 Type III. The bag shall be weighed empty; and then filled with 2 quarts of water and closed and weighed. The closed bag shall be weighed to the nearest gram. The bag shall be placed on a flat, horizontal supporting surface. A flat board shall be placed on top of the bag, but not on the neck. A load of 300 pounds shall be applied onto the board, using weights or a press. The board and the flat supporting surface shall be kept approximately parallel. The load shall be maintained for one hour, and then the bag and contents shall be weighed.

4.4.7 Load strength of tab. The type III bag shall be filled with 2 quarts of water and closed. The bag shall be suspended by the eye of the tab upside down on a hook. The bag shall be suspended for one hour.

4.4.8 Leakage. The type III bag shall be filled with 2 quarts of water and closed. All traces of water on the exterior of the bag, including the opening, shall be removed. The bag shall be placed on a flat, horizontal, supporting surface. A flat board shall be placed on top of the bag, but not on the neck. A load of 150 pounds shall be applied onto the board, using weights or a press. The board and the flat supporting surface shall be kept approximately parallel. The load shall be maintained for one hour, and the bag shall be examined for evidence of leakage.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (See 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. One bag of the type specified shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Intermediate package. Six unit packages containing type II or type III bags shall be packaged in a box conforming to PP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification. An intermediate package shall not be required for type I bags.

5.1.2 Level C. Bags shall be clean dry and individually packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Sixty unit packages containing type I bags or sixteen intermediate packages (96 units) containing type II bags or twelve intermediate packages (72 units) containing size III bags shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, waterproofing and banding shall be as specified in the appendix of the box specification.

5.2.2 Level B. Sixty unit packages containing type I bags or sixteen intermediate packages (96 units) containing type II bags or twelve intermediate packages (72 units) containing size III bags shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged bags of one type only shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

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5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bags are intended for use in hospitals, and other medical facilities.

6.2 Ordering data. Purchasers should select the preferred options herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection and applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Type
6530-00-770-7000	BAG, ICE, Oval	I
6530-00-770-7500	BAG, ICE, Throat Type	II
6530-00-770-6425	BAG, HOT WATER-ICE, Stopperless	III

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6.4 Supersession data. This specification supersedes the following documents:

Int. Fed. Spec.
ZZ-B-00586C(DSA-DM)
dated January 13, 1971
Fed. Spec. ZZ-B-586A,
dated August 1, 1944
Int. Fed. Spec.
ZZ-B-0045B(DSA-DM)
dated August 24, 1971
Fed. Spec. ZZ-B-45,
dated September 24, 1957
Int. Fed. Spec.
ZZ-B-0050B(DSA-DM)
dated January 10, 1970
Fed. Spec. ZZ-B-50,
dated June 28, 1955

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Civil Agency Coordinating Activities:
GSA-FSS
VA-DMS

Project No. 6530-1413

U. S. GOVERNMENT PRINTING OFFICE : 1977 - 241-233/1128

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DOCUMENT IDENTIFIER AND TITLE
ZZ-B-586D - BAGS, ICE AND HOT WATER

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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